

Project 1: E-Commerce Customer & Operations Analytics

Dataset: Olist Brazilian E-Commerce Public Dataset

Dataset link: <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

Business scenario: You are a Data Analyst at an online marketplace operating across Brazil. Management wants to understand who its customers are, what drives purchasing behavior and satisfaction, and where operational problems are occurring.

Business problem: Build an end-to-end analytics solution that explains sales performance, customer behavior, delivery performance, customer satisfaction and high-value customer segments, and supports operational decision-making.

EDA & Business Analysis

- Analyze orders, revenue, average order value, product categories, states/cities and freight cost.
- Study repeat vs. one-time customers and identify high-value customer locations.
- Measure late deliveries and investigate sellers, categories and geographies associated with delays.
- Analyze the relationship between delivery delay, freight cost and customer review score.

Customer Segmentation

- Build customer-level features using Recency, Frequency, Monetary value, average order value, average review score, average freight cost, delivery delay and category diversity.
- Apply K-Means clustering and select K using the Elbow method and Silhouette score.
- Profile segments such as premium loyal customers, high-value at-risk customers, occasional customers and new/low-engagement customers.
- Translate each segment into a business action such as retention, loyalty or re-engagement.

Predictive Analytics

- Primary prediction problem: predict whether an order will be delivered late.
- Target: Late = 1 when Actual Delivery Date > Estimated Delivery Date, otherwise 0.
- Candidate models: Logistic Regression, Decision Tree, Random Forest and Gradient Boosting/XGBoost.
- Evaluate with Precision, Recall, F1-score, ROC-AUC and a Confusion Matrix.
- Optional extension: predict poor customer satisfaction using review_score <= 2.

Dashboarding

- Executive Overview: revenue, orders, customers, average order value, repeat-customer rate, review score and late-delivery rate.
- Customer Analytics: customer segments, segment revenue, spending and satisfaction.
- Operations & Delivery: late vs. on-time orders, delay distribution, seller/category/geography risk.
- Predictive Analytics: model performance, feature importance and high-risk orders.

Project 2: Bank Customer & Campaign Analytics

Dataset: UCI Bank Marketing Dataset

Dataset link: <https://archive.ics.uci.edu/dataset/222/bank+marketing>

Business scenario: You are a Data Analyst at a retail bank running outbound campaigns to encourage customers to subscribe to a term deposit. The bank wants to improve targeting and reduce ineffective contacts.

Business problem: Understand the customer base, identify meaningful customer segments, predict campaign response and decide which customers should be prioritized for future campaigns.

EDA & Campaign Analysis

- Analyze age, occupation, education, balance, housing loan and personal loan patterns.
- Measure campaign conversion rate and compare outcomes by contact method, month and previous campaign outcome.
- Study the relationship between campaign contacts, customer characteristics and subscription success.
- Identify where marketing effort is being spent without strong conversion.

Customer Segmentation

- Create customer features from age, balance, campaign contacts, previous contacts, loan status and previous campaign outcome.
- Apply K-Means clustering and evaluate K using Elbow and Silhouette methods.
- Create interpretable customer personas such as high-value customers, young growth customers, credit-dependent customers, highly responsive customers and low-engagement customers.

Predictive Analytics & Gradient Descent

- Predict whether a customer will subscribe to the term deposit: $y = \text{yes/no}$.
- Use Logistic Regression as the primary model so the mechanics of gradient descent can be taught directly.
- Model: linear combination $\rightarrow$ sigmoid probability $\rightarrow$ log loss $\rightarrow$ gradient $\rightarrow$ gradient descent $\rightarrow$ updated weights.
- Compare Logistic Regression with Decision Tree, Random Forest and Gradient Boosting/XGBoost.
- Discuss why Logistic Regression uses gradient descent while a Decision Tree does not, and how Gradient Boosting uses gradient-based optimization.
- Interpret coefficients/feature importance and discuss data leakage. In particular, duration is highly predictive but is only known after the call, so it should not be used for a pre-call targeting model.

Campaign Optimization

- Assume the bank can contact only 10,000 customers next month.
- Rank customers by predicted subscription probability and estimate expected conversions and campaign efficiency.
- Turn the model into a business decision: who should be contacted first?

Dashboarding

- Executive Dashboard: customers, contacts, conversion rate, average balance and successful subscriptions.
- Customer Segmentation: segment size, balance, conversion and demographic profile.
- Campaign Analytics: conversion by contact method, month, contacts and previous outcome.
- ML Prediction: response probability, high-propensity customers, important features and model performance.

Project 3: Marketing Campaign Optimization

Dataset: Maven Marketing Campaign Results

Dataset link: <https://mavenanalytics.io/data-playground/marketing-campaign-results>

Business scenario: A consumer company is spending heavily on marketing campaigns but does not know which customers to target, which channels to use or which customer segments are most valuable.

Business problem: Analyze historical customer and campaign data to improve marketing targeting, customer understanding and campaign effectiveness.

EDA & Marketing Analysis

- Profile customers using age, education, marital status, income, household composition and country.
- Analyze spending across products and purchases through web, catalog and store channels.
- Measure campaign acceptance and compare the performance of campaigns and channels.
- Study relationships among income, spending, age, product preference, website visits and campaign response.

Customer Segmentation

- Create features from income, total spending, recency, web/catalog/store purchases, website visits and campaign responses.
- Apply K-Means clustering with Elbow and Silhouette validation.
- Interpret segments such as high-value omnichannel customers, digital customers, store-oriented customers, low-engagement customers and campaign-responsive customers.

Predictive Analytics & Gradient Descent

- Create a target such as whether a customer will respond to the next marketing campaign.
- Use Logistic Regression and teach gradient descent through the log-loss objective and iterative weight updates.
- Compare against Decision Tree, Random Forest and Gradient Boosting.
- Evaluate classification performance and interpret the features associated with higher response probability.
- Explicitly discuss leakage: only information available at the time of campaign targeting should be used as model inputs.

Dashboarding

- Marketing Overview: customers, spending, campaign acceptance, average income and channel purchases.
- Customer Segmentation: segment size, spend, response and demographic profile.
- Campaign Analytics: campaign-wise acceptance, channel performance and response by segment.
- Predictive Analytics: response probability, high-propensity customers and important predictive features.

Project 4: Multimodal Product & Customer Analytics

Dataset: Amazon Reviews'23 - McAuley Lab

Dataset link: <https://huggingface.co/datasets/McAuley-Lab/Amazon-Reviews-2023>

Dataset scope: The dataset combines review text, customer-uploaded images, ratings, helpful votes, timestamps, product metadata, category and brand information. The full 2023 release is very large, so a single category should be selected for a classroom project.

Business scenario: Build a multimodal product and customer analytics system for an e-commerce marketplace using text, images and structured product/review information together.

Business problem: Understand product performance and customer sentiment, identify product/customer groups and build predictive models that benefit from combining multiple modalities.

EDA & Analytics

- Analyze ratings, prices, categories, products, review volumes and customer engagement.
- Explore text characteristics, sentiment, recurring complaint themes and helpfulness.
- Examine image availability and basic visual characteristics of products/reviews.
- Study relationships between product attributes, ratings, review text and customer behavior.

Clustering

- Customer segmentation using review/purchase behavior features.
- Product segmentation using text embeddings, image embeddings and structured product attributes.
- Compare clusters and interpret them as meaningful customer or product personas.

Predictive Analytics

- Predict review sentiment or rating from available information.
- Predict whether a review is likely to receive high helpfulness.
- Optional business extension: predict customer engagement or product demand using multimodal features.
- Compare structured-only models with text-only, image-only and multimodal models.

Multimodal Modeling

- Image -> image embedding.
- Review text -> text embedding.
- Structured fields -> numerical/categorical features.
- Combine the representations and train a prediction model.
- Evaluate whether combining modalities actually improves performance over single-modality baselines.

Dashboarding

- Product Overview: ratings, review volume, price, category and brand performance.
- Customer/Review Analytics: sentiment, helpfulness and recurring themes.
- Product Segmentation: cluster size and representative products.
- Prediction: model performance and examples of multimodal predictions.